Cover Page

HACK FOR SOCIAL CAUSE | HSC-27 HACKATHON SUBMISSION

BlueHawk Civic Nexus: Autonomous Edge-AI Street Rover & GIS Urban Governance Grid for Proactive Emergency Threat Alerting and Sustainable Municipal Action

Problem Statement / Concept Title	**BlueHawk Civic Nexus** - H.A.W.K. Edge-AI Street Rover + BlueBridge GIS Urban Governance Grid
Theme	**Theme 4:** Governance & Civic Technology | **Theme 7:** AI, GIS & Emerging Technologies
Team Name (Registered on portal)	**BlueHawk Civic Nexus**
Team ID (Registered on portal)	**HSC|UP|00287**
State / UT	**Uttar Pradesh**

ALIGNED WITH UN SUSTAINABLE DEVELOPMENT GOALS

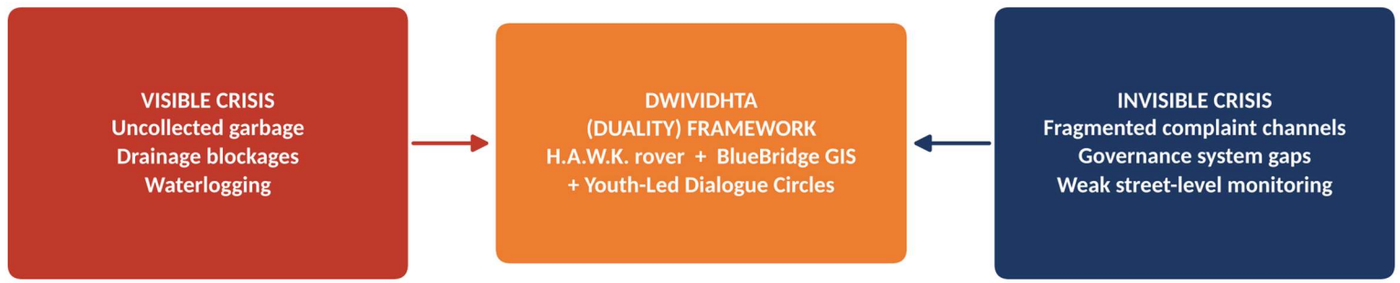
SDG 3	SDG 9	SDG 11	SDG 16
Good Health & Well-being	Industry, Innovation & Infrastructure	Sustainable Cities & Communities	Peace, Justice & Strong Institutions

Lead Innovator: Aravind K. T., B.Tech Electrical Engineering (Class of 2028), IIT Kanpur | Live App: https://bluehawk-civic-nexus.streamlit.app

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Problem – Proposed Solution

THE DWIVIDHTA (DUALITY) FRAMEWORK: closing the gap between what the city shows and what the system misses



CORE PROBLEM

- **Dual visibility crisis** in urban and semi-urban wards
- **Visible:** uncollected garbage, drainage blockages, waterlogging
- **Invisible:** governance gaps and weak street-level security monitoring in vulnerable corridors
- Complaints scattered across **unlinked channels**

PROPOSED SOLUTION

- **Closed-loop hybrid ecosystem** with two linked halves
- **H.A.W.K.:** low-cost autonomous street patrol rover
- **BlueBridge:** AI/NLP classification + PostgreSQL/PostGIS spatial GIS grid
- One pipeline: detect, classify, locate, route, act, learn

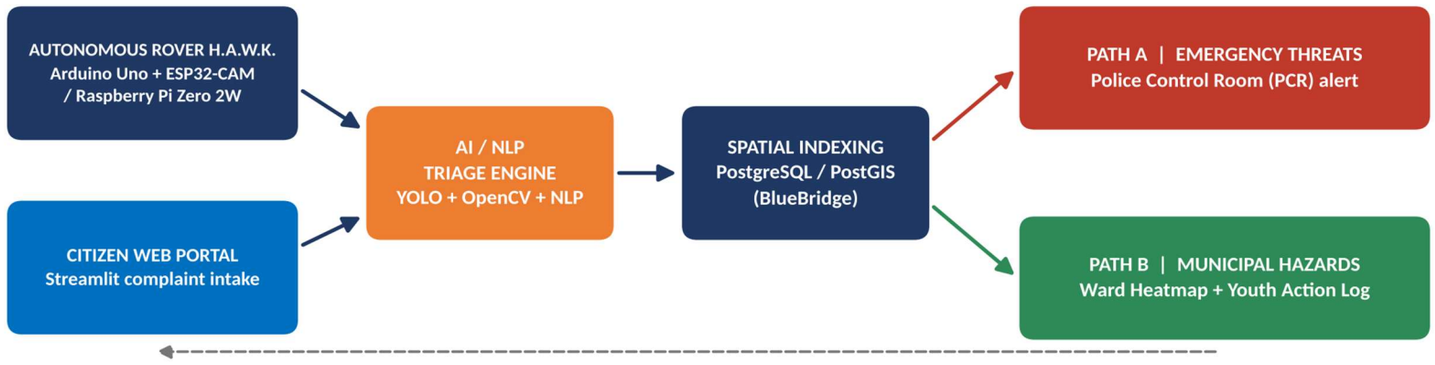
INNOVATION & UNIQUENESS

- Bridges **embedded robotics** with **cloud GIS analytics**
- Splits every alert into **Path A** (emergency) or **Path B** (municipal)
- Pairs technical precision with **community empathy** via Youth-Led Peace & Dialogue Circles
- Low-cost, modular, open-source

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Technical Approach

CLOSED-LOOP DATA FLOW: Ingest > Triage > Index > Route > Act > Feedback



HARDWARE STACK

- Arduino Uno R3 + Adafruit Motor Shield
- HC-SR04 ultrasonic, IR line/proximity, UV sensors
- Metal-gear servos
- Edge vision nodes: ESP32-CAM / Raspberry Pi Zero 2W

SOFTWARE & AI STACK

- Python: Streamlit, Pandas, NumPy
- OpenCV + YOLO (vision); NLP models (text triage)
- PostgreSQL / PostGIS spatial database
- Open-source throughout

LIVE LINKS & AI DISCLOSURE

GitHub: https://github.com/kt2141/BlueHawk-Civic-Nexus

Live App: https://bluehawk-civic-nexus.streamlit.app

Demo Video: Uploaded directly into the My bharat Portal

AI usage: YOLO/OpenCV + NLP power the core classifier. Generative AI assisted ~15-20% (formatting only). Hardware design, Arduino firmware and governance frameworks are 100% original.

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Impact Potential

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25-40%	2 PATHS	1 GRID	5
Projected faster cleanup & emergency response	Emergency vs municipal routing, automatic	All complaints unified on one PostGIS map	Stakeholder groups connected

IMPACT ON TARGET AUDIENCE

- **Residents:** faster fixes, one place to report, visible follow-through
- **Vulnerable corridors:** proactive patrol coverage and earlier threat alerts
- **Quantitative:** 25-40% faster cleanup and emergency response via spatial routing and automated hotspot prioritization (projected target, to be validated in pilot)

BENEFITS

- **Social:** enhanced public safety; structured youth empowerment
- **Health:** fewer garbage, drainage and waterlogging hazards
- **Governance:** higher administrative transparency and accountability
- **Economic:** low-cost hardware; crews dispatched where need is highest

STAKEHOLDERS

- **Citizens:** report and track
- **Municipal Workers:** prioritized, mapped tasks
- **Police Control Rooms:** verified emergency alerts
- **Youth Volunteers:** Action Log and Dialogue Circles
- **Urban Policymakers:** ward-level heatmap evidence

Dimension	Today (fragmented)	With BlueHawk Civic Nexus
Complaint handling	Unlinked channels, no common map	One geotagged queue on the PostGIS grid
Emergency alerts	Manual, delayed street-level monitoring	AI-triaged Path A alert straight to the PCR
Cleanup planning	Reactive, complaint by complaint	Hotspot-prioritized routing from the ward heatmap
Youth role	Ad hoc volunteering	Structured Youth Action Log + Dialogue Circles

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Feasibility and Risk Analysis

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LOW COST	OPEN SOURCE	WORKING	PHASED
Arduino / ESP32 modular parts	PostGIS + Python stack	Live app + public GitHub repo	Managed corridors first

FEASIBILITY ANALYSIS: HIGH VIABILITY

- **Technical:** modular, low-cost parts (Arduino / ESP32) - each module can be built, tested and replaced independently
- **Software:** open-source PostGIS architecture, no licence cost; live Streamlit app and public GitHub repository already available
- **Deployment:** phased - managed campus / ward corridors first, then wider wards

POTENTIAL CHALLENGES, RISKS & STRATEGIES

Risk / Challenge	Level	Mitigation Strategy
Terrain obstacles and hardware wear	Medium	Restrict initial deployments to managed campus / ward corridors; lightweight plastic/rubber wheel configuration; scheduled maintenance
Public adoption friction	Medium	Pair rover patrols with Youth-Led Peace & Dialogue Circles to build community trust and transparency
False alarms on emergency path	Medium	Confidence thresholds in the AI triage engine; human verification at the Police Control Room before dispatch
Privacy and data protection	Medium	Process on the edge node, store only geotagged event records, role-based access to the GIS grid

PHASED ROLLOUT

PHASE 1: PILOT	PHASE 2: WARD EXPANSION	PHASE 3: CITY INTEGRATION
Managed campus / ward corridors; validate detection accuracy and response-time gains	More rovers and corridors; Youth Action Log and Dialogue Circles scaled up	Link with municipal systems and Police Control Room workflows

Team Name
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Team

NATIONAL AWARD	UN GENEVA	NTU SINGAPORE	PARIS 2026
Devbhoomi Rashtriya Ratna Puraskar 2025	UN Social Forum 2025 delegate	i-SEA Fellow 2026	Global Youth Festival delegate

TEAM COMPOSITION & ROLES

Aravind K. T. - Lead Innovator

B.Tech Electrical Engineering (Class of 2028), IIT Kanpur, Hall of Residence 5; AISHE Code U-0517

- Robotics hardware & Arduino firmware
- AI/NLP triage and PostGIS / Streamlit platform
- Governance framework and youth-engagement model

Leadership: Former Secretary, Public Policy & Opinion Cell (IIT Kanpur); Founder, Youth-Led Peace & Dialogue Circles (Karnataka); UPG Sustainability Champion; UNESCO APCEIU GCED Certified; Official ICSC Ambassador

CHALLENGES OVERCOME

- **Weight and traction:** moved from heavy mild-steel wheels to optimized plastic/rubber configurations
- **Thermal servo loads:** diagnosed and resolved overheating under load
- **Integration:** united embedded hardware, cloud GIS and AI into one closed-loop pipeline
- **Trust:** linked technology to grassroots dialogue, not just dashboards

JOURNEY & MILESTONES

- **Devbhoomi Rashtriya Ratna Puraskar 2025:** National Award for Outstanding Contribution to Youth Welfare
- **Youth Leadership Outstanding Awardee:** Top 50 Youth Leaders in Asia-Pacific, Seoul
- **UN Social Forum 2025:** Officially accredited delegate, Palais des Nations, Geneva
- **i-SEA Fellow 2026:** Nanyang Technological University (NTU), Singapore
- **My Bharat Budget Quest 2026:** National Finalist, top among 13 lakh+ participants
- **Global Youth Festival 2026, Paris:** Fully funded delegate selection

Team Name
BlueHawk Civic Nexus

References

Reference	Details / Link
Digital Portfolio (Aravind K. T.)	https://sites.google.com/view/aravind-k-t-iitk/
Open-Source Code (GitHub)	https://github.com/kt2141/BlueHawk-Civic-Nexus
Live Deployment (Streamlit)	https://bluehawk-civic-nexus.streamlit.app
Institutional Guidelines	Ministry of Youth Affairs & Sports (MoYAS) and IIT Bombay VBYLD 2027 Framework
UN Sustainable Development Goals 3, 9, 11, 16	https://sdgs.un.org/goals
Technology documentation	PostGIS (postgis.net) | OpenCV (opencv.org) | Streamlit (streamlit.io) | Ultralytics YOLO (docs.ultralytics.com) | Arduino (arduino.cc)

VERIFICATION SUMMARY

Innovator: Aravind K. T., IIT Kanpur (AISHE Code U-0517) | **Team:** BlueHawk Civic Nexus | **State:** Uttar Pradesh | **Themes:** 4 and 7 | **SDGs:** 3, 9, 11, 16
Originality: hardware design, Arduino firmware and governance frameworks are 100% original; generative AI assisted ~15-20% of formatting only.